Research statement: A statistical study of topological and geometric descriptors

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Modern machine learning problems require the treatment of datasets which are more and more intricate, with large size and nonlinear structure. During my Ph.D. thesis, I studied methods from **Topological Data Analysis** (or TDA), which is a recent field at the intersection of machine learning, computational geometry, and topology. TDA aims at developing new techniques to extract key topological and geometric features from complex datasets, with successes in various scientific fields, ranging from biology [YSH+09], to chemistry [NHH+15] or material science [LBD+17]. In the two first sections of this document, I detail my contributions to two different approaches used in TDA, namely manifold inference and persistent homology theory, with an emphasis put on **understanding the statistical properties** of the different objects encountered. The last section is dedicated to future projects, especially focusing on **statistical applications of optimal transport theory**. Indeed, I believe that optimal transport theory gives a fruitful framework to study the non-linearity of complex datasets in an efficient and meaningful way.

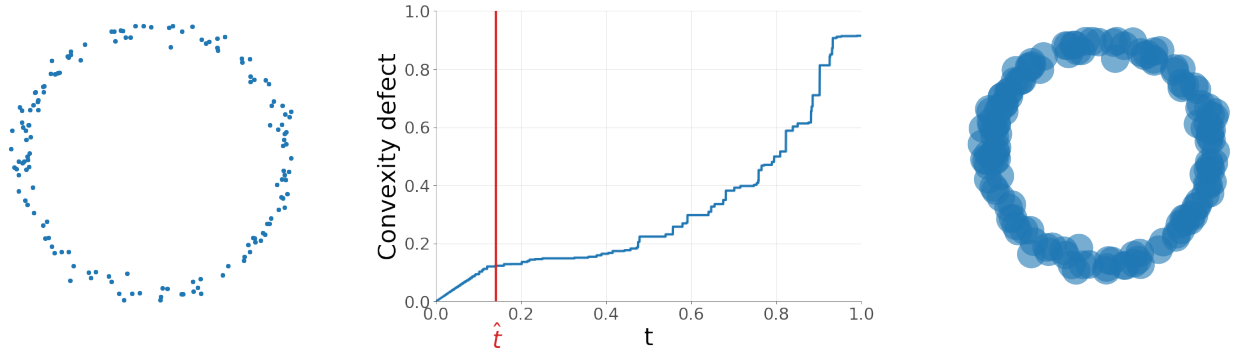
Minimax estimation in manifold inference

The first class of problems I have studied broadly falls under the term of "manifold inference". Such problems all make the key assumption that some observed dataset $\mathbb{X}_n = \{X_1, \dots, X_n\}$, which lies in a possibly high-dimensional space $\mathbb{R}^D$, is actually close to some low-dimensional shape M (which is modeled as a d-dimensional manifold with $d < D$). This manifold assumption generalizes common hypotheses made in high-dimensional statistics, such as sparsity of the dataset, the single index model, or the presence of a low-dimensional linear subspace which fits the dataset. A program which has been fruitfully developed over the past decade consists in using $\mathbb{X}_n$ to design a variety of descriptors that help to **understand geometric features of the underlying shape** M, such as its dimension [KRW19], its homology groups [NSW08], its curvature [AKC+19], etc. From a statistical perspective, the question of the optimality of those estimators is then raised. Here, optimality has to be understood in a minimax sense: if $\mathcal{P}$ is a set of probability distributions on $\mathbb{R}^D$, and $\theta : \mathcal{P} \to (E, d)$ is a functional of interest (taking its values in some metric space (E, d)), then the minimax risk of estimation of θ on the statistical model $\mathcal{P}$ given n observations is equal to

$$\mathcal{R}_n(\theta; \mathcal{P}; d) := \inf_{\hat{\theta}_n} \sup_{\mu \in \mathcal{P}} \mathbb{E}_{\mu^{\otimes n}}[d(\hat{\theta}_n, \theta(\mu))], \tag{1}$$

where the infimum is taken over all estimators $\hat{\theta}_n : (\mathbb{R}^D)^n \to E$. To put it another way, the minimax risk is the best risk an estimator can attain uniformly on the class $\mathcal{P}$.

Adaptive manifold estimation A first issue consists in designing statistical models $\mathcal{P}$ that encapsulate the regularity of the underlying manifold M. The key quantity to do so is the reach of a manifold $\tau(M)$, given by the distance between the manifold and its medial axis. Introduced in [Fed59], it captures both the local curvature of M and its global regularity. The model $\mathcal{P}$ is then the set of all laws μ supported on a d-dimensional manifold M with $\tau(M) \geq \tau_{\min}$, having a density f on M (which is upper bounded and lower bounded by a positive constant). If we consider the problem of the reconstruction of the manifold M for the



(a) Input data $\mathbb{X}_n$

(b) Convexity defect function

(c) Offset of the input data $\mathbb{X}_n$ with parameter $\hat{t}$

Figure 1: Computing the convexity defect function of a point cloud $\mathbb{X}_n$ allows the selection a good scale $\hat{t}$ at which one should look at the dataset.

Hausdorff distance d_H, the minimax risk on those models is known to be of order (see [GPPV$^+$12, KZ$^+$15])

$$\mathcal{R}_n(M; \mathcal{P}, d_H) \asymp \left(\frac{\ln n}{n}\right)^{2/d}, \tag{2}$$

while several minimax estimators have been proposed in the litterature [AL18, GPPV$^+$12]. Those estimators all depend on several hyperparameters. Those hyperparameters have to be properly tuned with respect to the parameters defining the model in order for the estimators to be minimax. In practice, those parameters are unknown, and the tuning of the hyperparameters turns out to be a real issue.

The problem of the tuning of hyperparameters is not limited to the framework of manifold inference and is a classical issue in statistics. A standard way to overcome it consists in designing *adaptive* estimators, i.e. estimators which are simultaneously minimax on a large class of models. In a work submitted to the Electronic Journal of Statistics [Div20], I showed that **it is possible to design such an adaptive estimator for the problem of manifold estimation**. This was made possible by introducing a particularly simple manifold estimator $\hat{M}_t$, depending on a single index t, and defined as the union of simplexes on $\mathbb{X}_n$ of size roughly equal to t. After showing that this estimator is minimax for a certain choice t_{opt} (depending on unknown parameters such as $\tau_{\min}$), I developed a procedure to select a scale t almost as good as the optimal one. This is done by computing the so-called **convexity defect function** of $\mathbb{X}_n$ [ALS13], defined as $t \mapsto d_H(\hat{M}_t, \mathbb{X}_n)$. I have shown that this function exhibits a sharp change of behavior around a threshold value $t^*(\mathbb{X}_n)$, and that selecting a value t just above this threshold gives an estimator with almost the same risk as $\hat{M}_{t_{opt}}$, leading to an adaptive estimator. The selected value $\hat{t}$ has to be understood as an appropriate scale to look at the dataset. See also Figure 1 for an illustration of the procedure.

Density estimation on manifolds with optimal transport A second question I considered is the reconstruction of the law μ generating the observations itself. It is possible to show that such a reconstruction problem is degenerate from a minimax perspective for a large class of standard losses on the set of probability measures (total variation, Hellinger distance, or other f-divergences) while being tractable for optimal transport distances such as the Wasserstein distances W_r. Even in the case of densities on the cube $[-1,1]^d$, the study of the reconstruction of smooth measures in terms of Wasserstein losses is very recent, having only been considered so far in [WB19], where minimax rates are exhibited on models taking into account the regularity of some underlying density on the cube $[-1,1]^d$. Building upon recent results from [AL$^+$19] on the reconstruction of local parametrizations of $\mathcal{C}^k$ manifolds, I showed that it is possible to **adapt classical kernel density estimators to the case of manifold-supported observations $\mathbb{X}_n$, while retaining the same minimax rates of convergence as for densities on the cube**.

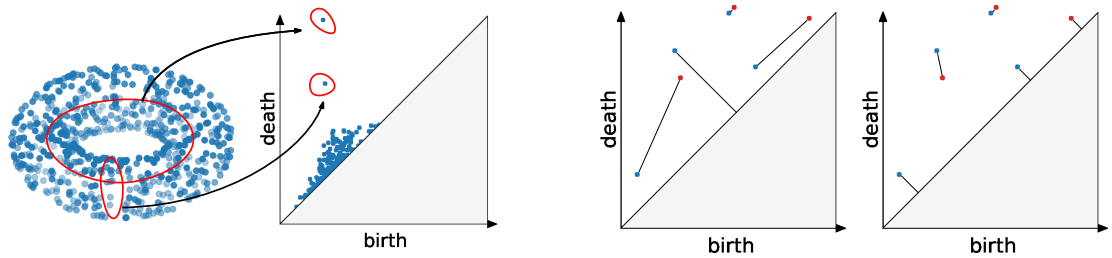


Figure 2: **Left display:** The persistence diagram of 700 points sampled on a torus. Each of the two distinguishable points in the diagram corresponds to a hole of the torus. **Right display:** Two matchings between a blue persistence diagram and a red persistence diagram. The rightmost matching is optimal, i.e. it attains the minimum in (3).

Understanding properties of random persistence diagrams

Persistent homology [EH10] is a popular approach in TDA. It provides efficient mathematical and algorithmic tools to understand the topology of some dataset (e.g. a graph or a an abstract metric space) by tracking the evolution of its homology at different scales. The algebraic machinery used in persistent homology is too complicated to be explained here, but a simple example is sufficient to understand the general idea. Consider $\mathbb{X}_n$ a set of points in $\mathbb{R}^D$, organized around a torus (see Figure 2). How can we formalize the presence of two "holes" in the dataset $\mathbb{X}_n$ (corresponding to the two holes of the torus)? A first idea consists in choosing an appropriate scale t, and look at the t offset of $\mathbb{X}_n$ (similarly to what was proposed in the first section) and then to look at the number of holes of the offset (i.e. its homology). An alternative perspective is to take a **multiscale approach**: we consider all possible values $t > 0$, and track how the number of holes of the t-offset varies as t grows. This process may be summarized by an object that we call a **persistence diagram**. It consists of a set a of points in the upper half-plane $\Omega := \{x = (t_1, t_2) \in \mathbb{R}^2,\ t_1 < t_2\}$, where each point (t_1, t_2) of a accounts for a hole which appeared in the process at the scale t_1, its birth date, and disappeared at the scale t_2, its death date. See Figure 2 for an illustration. In that sense, persistent homology is a **highly non-linear dimension reduction technique**, which takes as an input a set $\mathbb{X}_n$, and ouputs a two-dimensional dataset summarizing its topology.

Let $\mathcal{D}$ be the set of persistence diagrams. Due to algebraic considerations, the proximity between two persistence diagrams $a, b \in \mathcal{D}$ is usually measured thanks to d_p distances, defined as follows. Let $\Gamma(a, b)$ be the set of bijections (or matchings) between $a \cup \partial\Omega$ and $b \cup \partial\Omega$, where $\partial\Omega = \{(t, t),\ t \in \mathbb{R}\}$ is the diagonal. Then,

$$d_p(a, b) := \inf_{\gamma \in \Gamma(a,b)} \left(\sum_{x \in a \cup \partial\Omega} |x - \gamma(x)|^p \right)^{1/p}, \tag{3}$$

with usual modification if $p = \infty$ [EH10, Chapter VIII]. See also Figure 2.

The geometry of the space of persistence diagrams The corresponding metric space $(\mathcal{D}_p, d_p)$—where $\mathcal{D}_p = \{a \in \mathcal{D},\ d_p(a, 0) < \infty\}$—turns out to be highly non-linear, and deserves further examination. In a work made in collaboration with Théo Lacombe, published in the Journal of Applied and Computational Topology [DL20], we studied in detail this space by **interpreting the distance d_p as a partial optimal transport metric**. Taking such an approach has several benefits. First, it allows giving **a characterization of convergence with respect to the d_p distances in terms of weak convergence of measures**, which provides a very simple way to prove the continuity of applications on the space $\mathcal{D}_p$. Second, making this connection with optimal transport allows us to leverage standard tools from this field (see e.g. [LGL17]). We were in particular able to **prove the existence of barycenters for general distributions on the space $\mathcal{D}_p$**, generalizing existing results in the litterature [MMH11].

Representations of persistence diagrams and the choice of the weight function The nonlinear geometry of the metric space $(\mathcal{D}_p, d_p)$ makes the use of standard statistical tools rather awkward in the

space of persistence diagrams. As a workaround, feature maps $\Phi : \mathcal{D}_p \to \mathcal{B}$ (also called representations) are commonly used for the analysis, where $\mathcal{B}$ is a Banach space. A large class of feature maps, which we call linear, depends on some weight functions, the choice of which is a critical issue in practice. In a work with Wolfgang Polonik [DP19], published in the Journal of Applied and Computational Topology, we addressed this choice from a mathematical perspective, by considering an asymptotic setting. Assume that $\mathbb{X}_n$ is an i.i.d. sample from a density on $[0,1]^d$. Using tools from stochastic geometry, we were able to exactly characterize the weight functions for which the corresponding feature maps converge as n approaches infinity, and by doing so, we proved **laws of large numbers for the feature maps of such diagrams**. This theoretical result leads to a simple heuristic for tuning weight functions: **if the data lies near a d-dimensional manifold, then a sensible choice of weight function is the distance to the diagonal $\partial\Omega$ to the power α with $\alpha \geq d$.**

The expected value of a persistence diagram Instead of using feature maps, one could try to perform statistical tasks directly in the space $\mathcal{D}_p$ of persistence diagrams. It is for instance possible to define the expected value of a probability measure P on the space $\mathcal{D}_p$: this is the measure $\mathbb{E}_P[a]$ on Ω such that for all Borel sets $A \subset \Omega$,

$$\mathbb{E}_P[a](A) = \mathbb{E}_P[a(A)], \tag{4}$$

where $a \sim P$ and $a(A)$ is the number of points of the diagram in the region A. In a work made in collaboration with Fréderic Chazal [DC19], we showed that **the measure $\mathbb{E}_P[a]$ is a smooth object** (i.e. has a smooth density with respect to the Lebesgue measure) in a wide variety of settings, in particular when looking at persistence diagrams of random sets, or at the sublevel sets of the Brownian motion. The regularity of this measure can be leveraged to **perform kernel density estimation procedures** on a sample of persistence diagrams of law P, this method providing an object close to the underlying expected persistence diagram. This work was presented at SoCG 2018. It was then published in a special issue of the Journal of Computational Geometry ($\sim$ **top** 15% **of the papers of the conference**).

Further directions

As an instructor at the MIT, I will bring my experience in Topological Data Analysis, nonparametric statistics, geometric inference and optimal transport. My inclinations towards optimal transport theory and statistics would make a prospective collaboration with Prof. Philippe Rigollet particularly natural.

Bandwidth selection revisited with Wasserstein losses In [WB19], authors propose to approach a density with an adaptive estimator based on wavelet thresholding. This estimator is shown to be minimax adaptive for the Wasserstein loss W_r. However, its computation is a priori non-tractable, as it requires implementing some complicated projection functional. I believe that (i) a kernel density estimator also leads to a minimax estimator with respect to Wasserstein losses while being easily computable, and that (ii), it is possible to **design bandwidth selection procedures** in this context. Such selection procedures could for instance be based on modifications of the Goldenshluger-Lepski method, and have the potential to lead to new oracle inequalities. Such a project then raises natural questions, such as the comparison of the selected bandwidth to the bandwidth selected with more classical norms (such as the ℓ_2-norm).

Spectral clustering: can we do better? Spectral clustering consists in building an appropriate neighborhood graph on some dataset, and in then using the spectral properties of the Laplacian operator on the graph to partition the dataset into different clusters. It has been shown, in the case of datasets close to manifolds or to more general metric spaces [TGHS20], that such a procedure amounts to estimating the eigenvalues and the eigenvectors of the Laplace-Beltrami operator of the underlying space. This raises the question of the optimality of such a procedure: I conjecture that it is possible to **leverage the smoothness of the underlying signal generating the observations to estimate the spectrum of the Laplace-Beltrami operator with faster rates of convergence**. Due to similarities with kernel density estimation, I conjecture that such an enhanced estimator could be built by using an oscillating kernel to construct the neighborhood graph. The analysis of such an estimator could be conducted by studying precisely the spectral properties of convolution operators on metric spaces (similarly to [LM$^+$10]).

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